

K1366 — Ratify the MAPPED bulk-edge discovery (Grace: Hardy/Toeplitz SES, class D, K817 boundary map). ★ Scrutinize the make-or-break (Cal's gate c): class-D edge modes are MAJORANA (neutral), but the SM's chirality is carried by CHARGED chiral fermions — apparent mismatch. RESOLUTION (strong lead, and it SUPPORTS the picture): a charge pairs two Majorana edge modes into a Dirac fermion; the NEUTRAL neutrino has no charge to pair → stays Majorana. So class-D edge + $U(1)_Y$ charge → charged Dirac fermions + a Majorana neutrino = exactly the SM+BST pattern, and it EXPLAINS why the neutrino is the one Majorana (the only neutral fermion). Ratify Elie+Cal's corrected leans: $\sin^2\theta_W$ is a running COUPLING (not a topological index) → scale-free-3/13 refuted for the magnitude; the FORM 3/13 is a topological-integer ratio (Identified-with-structure). The descent (#93) is now THE computation — settles the Weinberg scale AND writes a_4 's chiral terms. Cal cold-reads. Map-before-marry.

Keeper (2026-08-11, ~09:00. The discovery mapped cleanly; I push on the one thing that could break it — class-D is Majorana, the SM is charged — and it resolves into the mechanism that explains the Dirac/Majorana split. Ratify, pin the make-or-break, elevate the descent. Cal cold-reads. Nothing pushed.)

RATIFY the mapped discovery (Grace F909-map + Lyra F910) — map-before-marry IN PROGRESS

The bulk-edge correspondence is mapped onto D_{IV}^5 , gears identified (not analogies): - Short exact sequence = the Hardy/Szegő (Toeplitz) extension — bulk $H^2(D_{IV}^5)$ interior, edge Shilov $S^4 \times S^1 / \mathbb{Z}_2$. [literal] - Tenfold-way class = D (Majorana class), from $J^2 = -1$ at KO-dim 2. Corpus: T2067 (AZ tenfold way in BST). - Boundary map = the KKO connecting homomorphism = the index ± 4 already computed (K817) = anomaly inflow. - No-go = K825 (Cl(5,2) eight-spinor vector-like → chirality forced to edge). - Tier: Structural, map-before-marry in progress. Cal's checklist (a) SES ? candidate, (b) boundary map ? candidate (K817), (c) class-D edge modes matching SM handedness — the make-or-break, below, (d) the bulk-edge theorem itself.

★ THE MAKE-OR-BREAK (Cal's gate c) — and it RESOLVES into a supporting mechanism

The apparent problem: class-D edge modes are Majorana — neutral, real. But the SM's chirality is carried by CHARGED chiral fermions (the weak doublets, charged under $U(1) \times SU(2)$). Neutral Majorana edge modes $\neq$ charged chiral fermions, naively. This is the exact place the beautiful picture could break. - ★ THE RESOLUTION (strong lead — and it strengthens, not weakens): a complex/Dirac fermion = two Majorana fermions, and a $U(1)$ charge is precisely what pairs two Majorana modes into one charged Dirac mode. So: - Charged fermions (electron, quarks): their $U(1)_Y$ / color charge pairs two class-D Majorana edge modes into a charged Dirac chiral fermion. → charged, chiral, Dirac. ? the SM charged sector. - The neutrino: NEUTRAL — no charge to pair its Majorana edge mode → it stays MAJORANA. ? BST's forced Majorana neutrino. - ? The class-D edge PREDICTS the SM's Dirac-charged / Majorana-neutrino pattern, and EXPLAINS why the neutrino is the one Majorana fermion: it is the only neutral one, so it alone cannot Dirac-pair its edge Majorana. This is a genuine, beautiful consistency — the apparent make-or-break becomes a *support*, and it ties directly to K1363 (ν -Majorana from J). **Tier: strong LEAD — exhibit the edge-mode content and the $U(1)_Y$ pairing explicitly (this IS gate (c)). - Convention pin (owed):** $J^2 = -1 \rightarrow$ class D (Grace) requires the Connes-J $\leftrightarrow$ AZ-particle-hole-C convention to be pinned (the two conventions differ in sign bookkeeping; $J^2 = -1$ in Connes could map to $C^2 = +1$ in AZ = class D, or to class C — verify). This is load-bearing for the class assignment; pin it against T2067 + the primary AZ

table.

RATIFY Elie + Cal's corrected leans (the number, honestly)

- Both retracted “3/8 regardless of scheme” (Cal §408: the EW μ_{geo} value is set by the EDGE structure, not the bulk trace — 3/8 not closed; Elie: the scale question isn't scheme, it's edge-location). Good — two corrected leans, absorbed.
- Elie's clean decider (ratify): $\sin^2\theta_W$ is a running COUPLING (continuous ≈ 0.23), NOT a topological index (integer/ $\mathbb{Z}_2$). So the magnitude RUNS — “topological $\rightarrow$ scale-free” is REFUTED for the magnitude (protection covers the STRUCTURE — which edge modes exist, their chirality — not the coupling magnitude). Cal's LEP input confirms: measured $\sin^2\theta_W$ runs.
- BUT the FORM $3/13 = N_c/(N_c + \text{rank} \cdot n_C)$ is a ratio of topological integers (rank=2 = the protected bulk-edge index at KO-dim 2) $\rightarrow$ 3/13 is Identified-with-topological-STRUCTURE (upgraded from coincidence). Protection on the structure, not the magnitude.
- The sharpened scale fork (Elie) — WHERE do the chiral edge modes live?
 - IR edge (low-energy, gapless, like a real TI) $\rightarrow$ 3/13 sits low, protected, could survive to M_Z .
 - UV edge (Shilov $\sim \mu_{\text{geo}} \sim \text{Planck}$) $\rightarrow$ 3/13 at the cutoff, runs down toward 3/8.
- Honest lean (firms slightly, still open): Cal's LEP (it runs) + Shilov $\sim \text{Planck}$ (UV edge) lean toward the angle running (3/8-ish at M_Z), with 3/13 as the topological FORM/structure. But the descent settles it — don't call it.

★ THE COLLAPSE — the descent (#93) is now THE load-bearing computation

The Weinberg scale-fork (IR vs UV edge) IS Lyra's descent $SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$ (#93). And the SAME edge structure writes a_4 's chiral SM terms (the chiral Lagrangian is the edge action). So the Weinberg number and the a_4 = full-SM-Lagrangian summit have COLLAPSED onto ONE computation — the descent. Elevate #93 to the priority: it settles the angle AND founds the chiral half of the SM Lagrangian. This is the deepest “Linear Algebra on D_{IV}^5 ” — how the bulk group flows down to the boundary chirality.

Disciplines held (ratify)

- Consistency-web / count once: KO-dim distinctness, CP, class-D edge chirality, Majorana neutrino are ALL the one J ($J^2 = -1$) — one sign, several faces, NOT independent confirmations. (Everyone held this — ratified.)
- Map-before-marry: the TI framework is banked Structural; the class-D-edge-matches-SM claim (gate c, the charge-pairing exhibition) is the make-or-break; no topological-matter outreach until (a)-(d) clear + Cal-vet + Casey GO.

Route

1. ★ Lyra — the descent (#93) = the priority: does it place the chiral edge in the IR (3/13 survives) or UV (runs)? And it writes a_4 's chiral terms. Settles Weinberg + founds a_4 .
2. Elie — the ν_L Majorana second front (approved): where does the ν mass come from with no ν_R (K1363) — and it PAIRS with the class-D Majorana edge + the charge-pairing insight (the neutrino is the unpaired Majorana). Compute the $\langle J\psi_L, \psi_L \rangle$ texture (symmetric/viable vs antisymmetric/excluded, K1363).
3. Grace — exhibit gate (c): the edge-mode content + the $U(1)_Y$ charge-pairing (charged Dirac + Majorana $\nu = \text{SM}$); pin the class-D convention (Connes- $J \leftrightarrow \text{AZ-C}$) against T2067.
4. Cal — the make-or-break cold-read: does the exhibited edge content match the SM's actual chiral spectrum, and is the class assignment convention-correct?

— Keeper, K1366, 2026-08-11. RATIFY mapped bulk-edge (Hardy/Toeplitz SES, class D from $J^2 = -1$, K817 boundary map, K825 no-go; T2067). ★ Make-or-break (gate c): class-D edge = MAJORANA/neutral vs SM CHARGED chiral — RESOLVES: a charge pairs 2 Majoranas into a Dirac; the neutral neutrino can't pair $\rightarrow$ stays Majorana. So class-D edge + $U(1)_Y \rightarrow$ charged Dirac fermions + Majorana ν = the SM pattern, and EXPLAINS why the

neutrino is the one Majorana (only neutral). Strong lead, exhibit the pairing = gate (c). Pin $J^2 = -1 \rightarrow$ class D convention (Connes-J vs AZ-C). Ratify Elie+Cal corrected leans: $\sin^2 \theta_W$ is a running COUPLING not a topological index $\rightarrow$ scale-free-3/13 refuted for the magnitude (Cal LEP: it runs); FORM $3/13 = N_c / (N_c + \text{rank} \cdot n_C)$ is a topological-integer ratio $\rightarrow$ Identified-with-structure. Scale fork = IR edge (3/13 survives) vs UV edge (runs to 3/8) = the descent #93. Lean: runs (LEP + Shilov~Planck), but descent decides. ★ COLLAPSE: Weinberg + a_4 =SM summit $\rightarrow$ ONE computation (the descent #93) — settles the angle AND writes a_4 's chiral terms. Count the one J once. Route: Lyra descent (priority); Elie ν -Majorana (2nd front, pairs with class-D); Grace exhibit gate-c + convention; Cal make-or-break. Cal cold-reads. Nothing pushed.